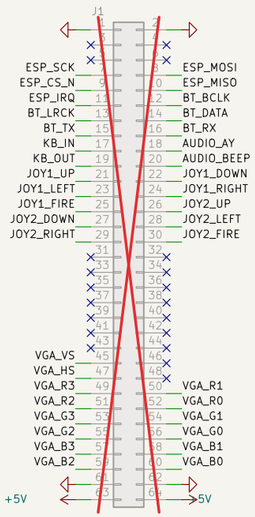
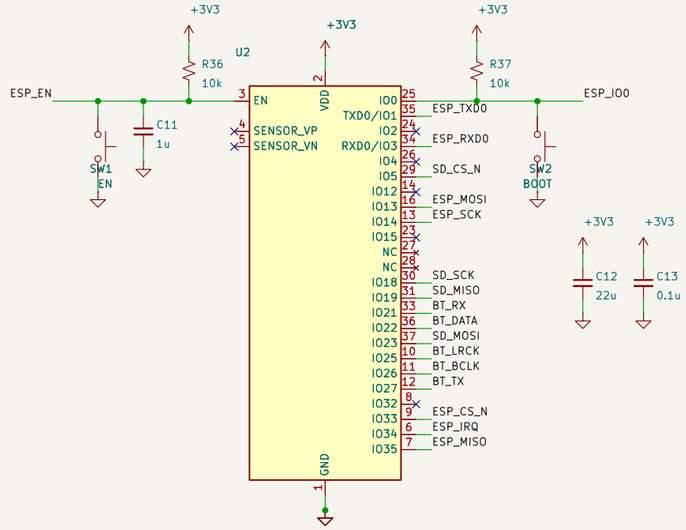
J1: host connector to the core board's U8 (pairs swapped)



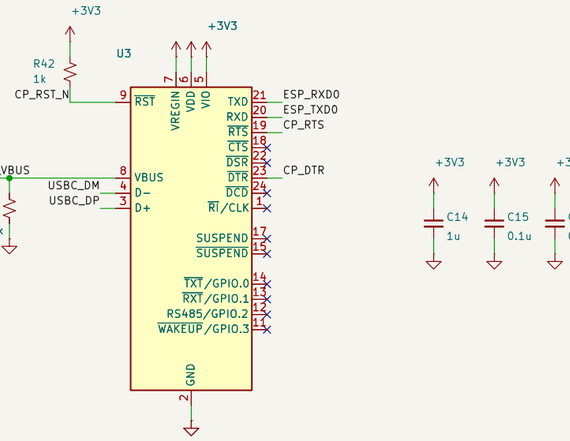
Power: USB-C 5 V -> SS34 -> +5V (core board via J1) -> LDO -> +3V3

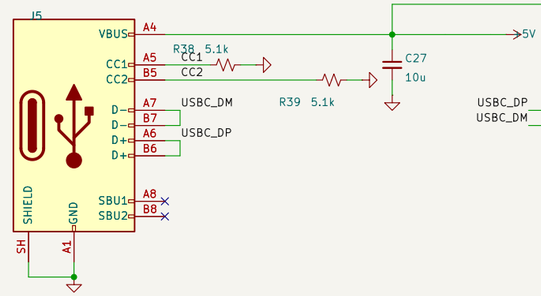
ESP32-WROVER-E-N16R8 (control UART on IO27 / IO21)

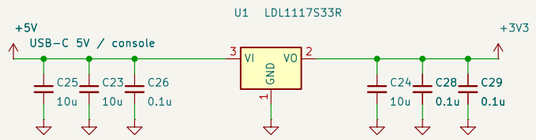


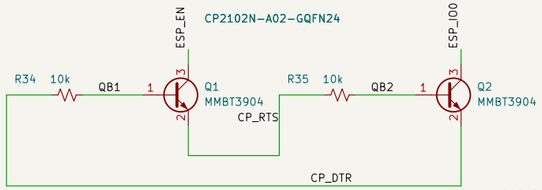
ESP32-WROVER-E-N16R8

CP2102N USB-UART on USB-C, DTR/RTS auto-reset

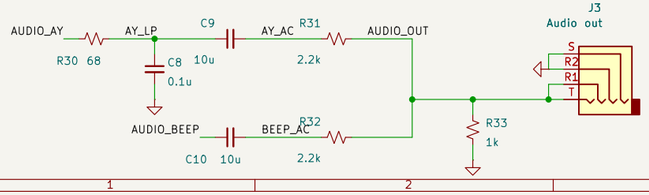


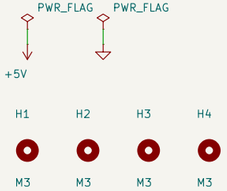




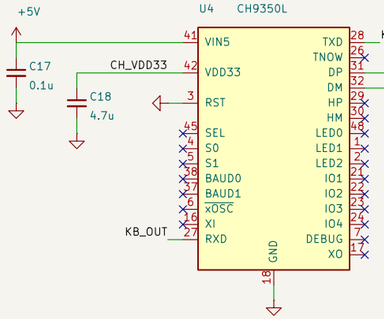


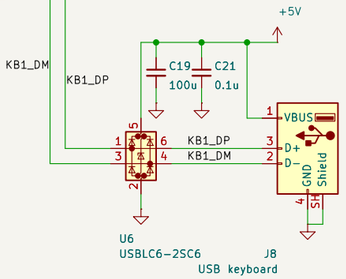
Audio line out (AY + beeper, mono on tip and ring)



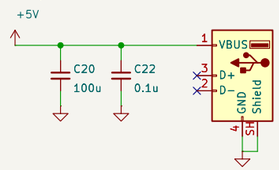


CH9350L: USB keyboard -> UART 115200

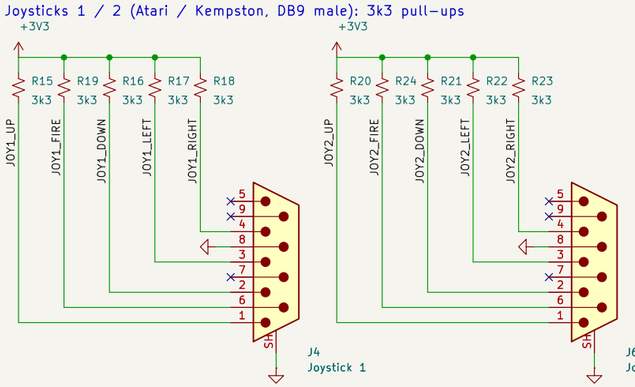




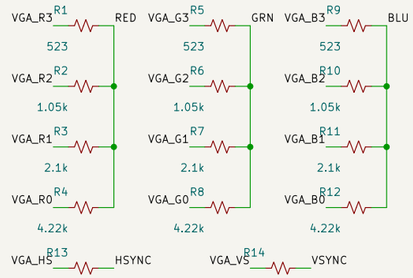
USB 2: 5 V only

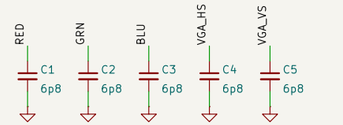


Joysticks 1 / 2 (Atari / Kempston, DB9 male): 3k3 pull-ups

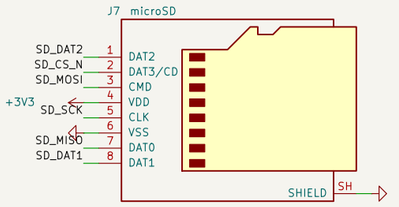


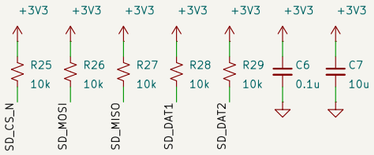
VGA: 4-bit DAC per colour





microSD (SPI, on the ESP32)





ESP32-WROVER-E, CP2102N, CH9350L, 2 USB-A, VGA, 2x DB9, SD, audio
Beside the QMTECH Cyclone IV core board, wired to its U8

Sheet: /
File: Extra_PCB_CycloneIV_R2.kicad_sch

Title: L I/O board (Extra_PCB_CycloneIV) revision 2

Size: B | Date: 2026-11-02 | **Rev: 2**
KiCad E.D.A. 10.0.1 | Id: 1/1